

Google Cloud

Cloud Day

Lisbon

RITA

Robô Inteligente de **T**riagem **A**utomatizada

Smart Automated Triage Agent



Who are we?

Proprietary + Confidential



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Overview



Challenge: smart triage

.....



Approach

.....



Demo

.....



Next Steps

.....

Emergency Services

are the convergence point
of all the **dysfunctions**
of the national health system.



**What's
the scenario?**

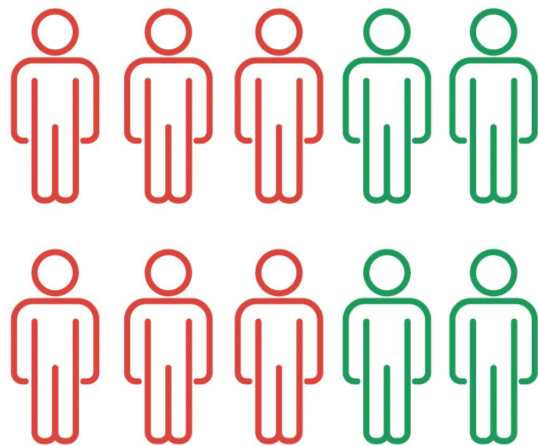
Shortage of professionals

-13 791
Nurses



Overcrowding

37,6%
**Non-emergent
care**



National data from 1st Jan. to 21st June 2024. Source: <https://www.sns.gov.pt/monitorizacao-do-sns/servicos-de-urgencia/> (accessed on 22/06/2024)

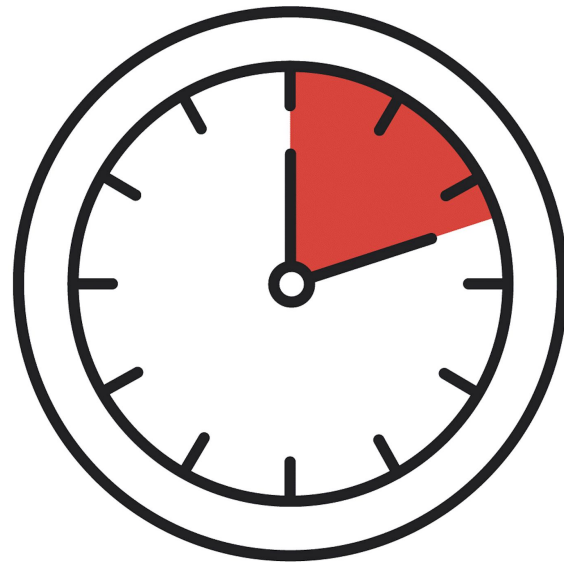
Wait time



20%

of total wait time*
corresponds to triage

*16 of 79 minutes since admission until first medical observation



National data from 1st Jan. to 21st June 2024. Source: <https://www.sns.gov.pt/monitorizacao-do-sns/servicos-de-urgencia/> (accessed on 22/06/2024)

An abstract graphic on the left side of the slide consisting of various green rounded rectangles and circles of different sizes, some overlapping and some outlined, creating a modern, layered look.

What can we do to improve it?

Provide better patient care
by combining
AI with **clinical expertise**.

Benefits & Impact

Social

Free healthcare professionals so that they can **focus on patient care**

Reduce non-emergent patient **influx** to the Emergency Service

Shorten on site **wait times**

vs. SNS 24 call

No call **queue**

vs. Avaliar Sintomas (SNS 24)

Natural language
input: text & speech

vs. both

Multilingual support

Benefits & Impact

Technical

Electronic health record (EHR) connection

Interoperability with other virtual and human **agents**

Continuous improvement of triage process thanks to **ML**

Infrastructure, **updates**, and security are managed by **Google**

Scalable and future-oriented



But...
HOW?

Manchester Triage System

Emergent

- Airway compromise
- Ineffective breathing
- Shock
- Unresponsive child
- Ongoing seizure

Very Urgent

- Uncontrolled major bleeding
- Repeated altered state of consciousness
- Very high temperature (fever)
- Severe pain
- Ongoing seizure

Urgent

- Minor bleeding
- Persistent vomiting
- High temperature (fever)
- Mild pain

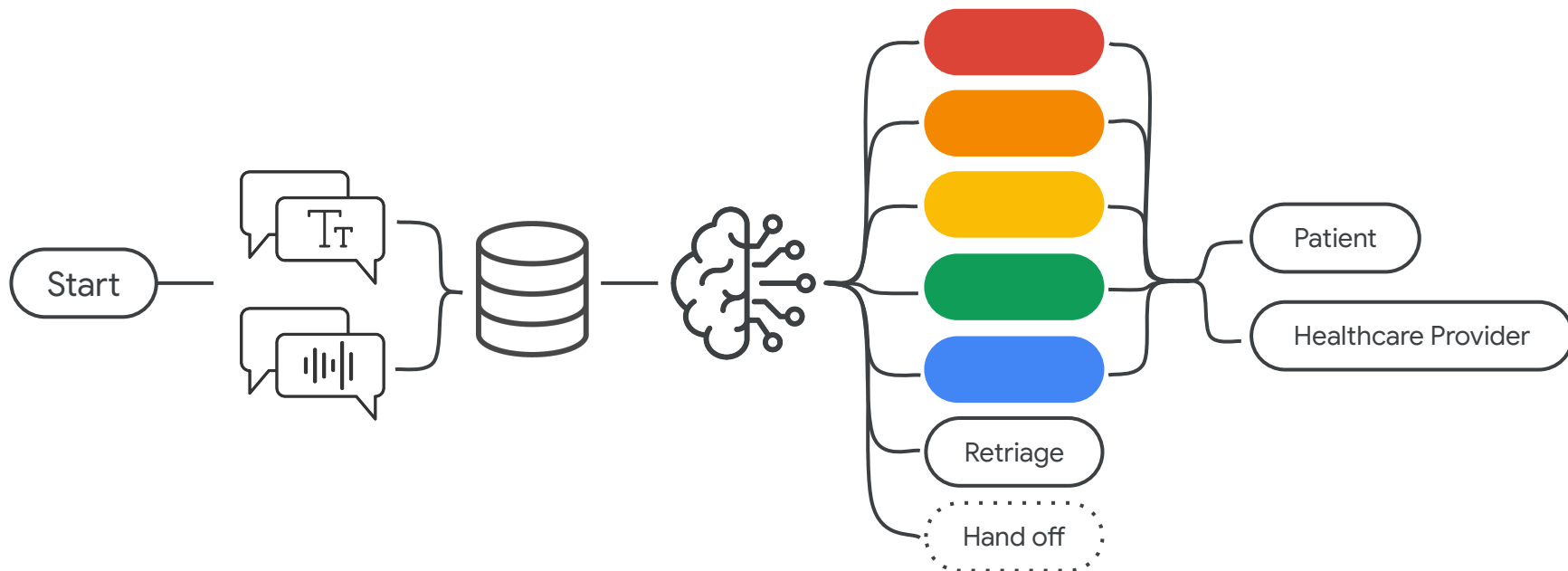
Not Very Urgent

- Subfebrile (low-grade fever)
- Vomiting
- Mild pain < 7 days
- Recent problem

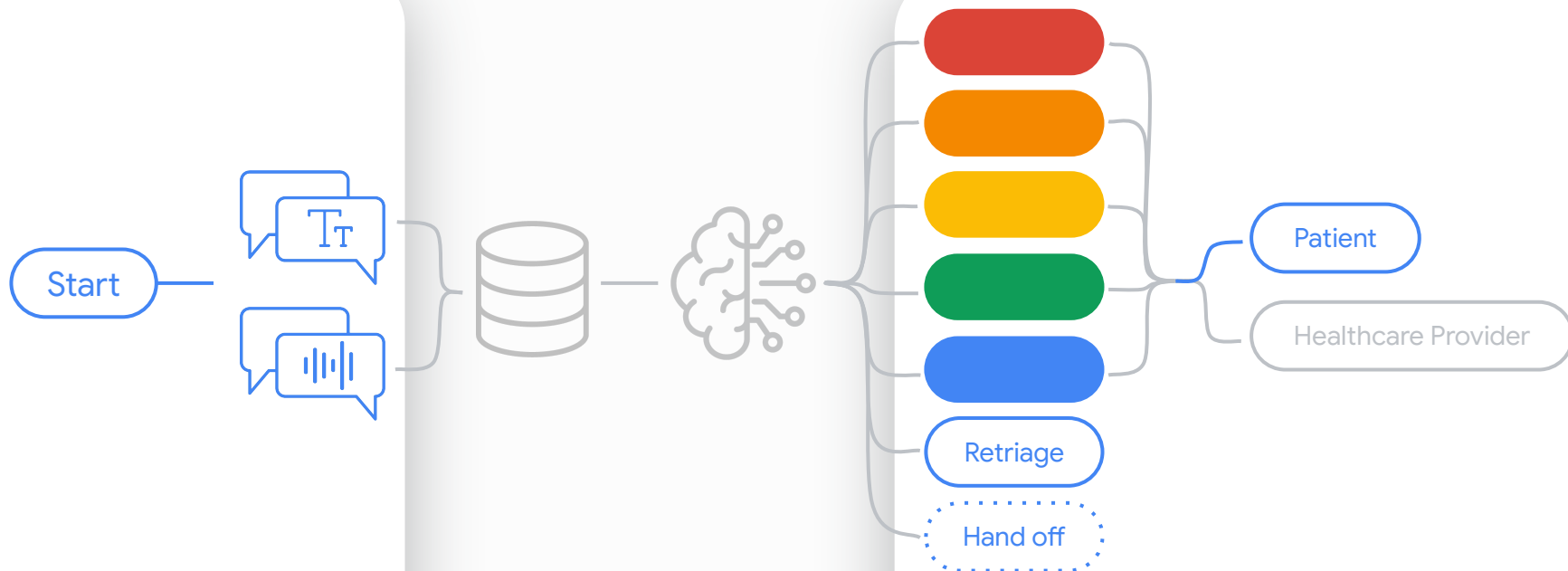
Not Urgent

None of the previous

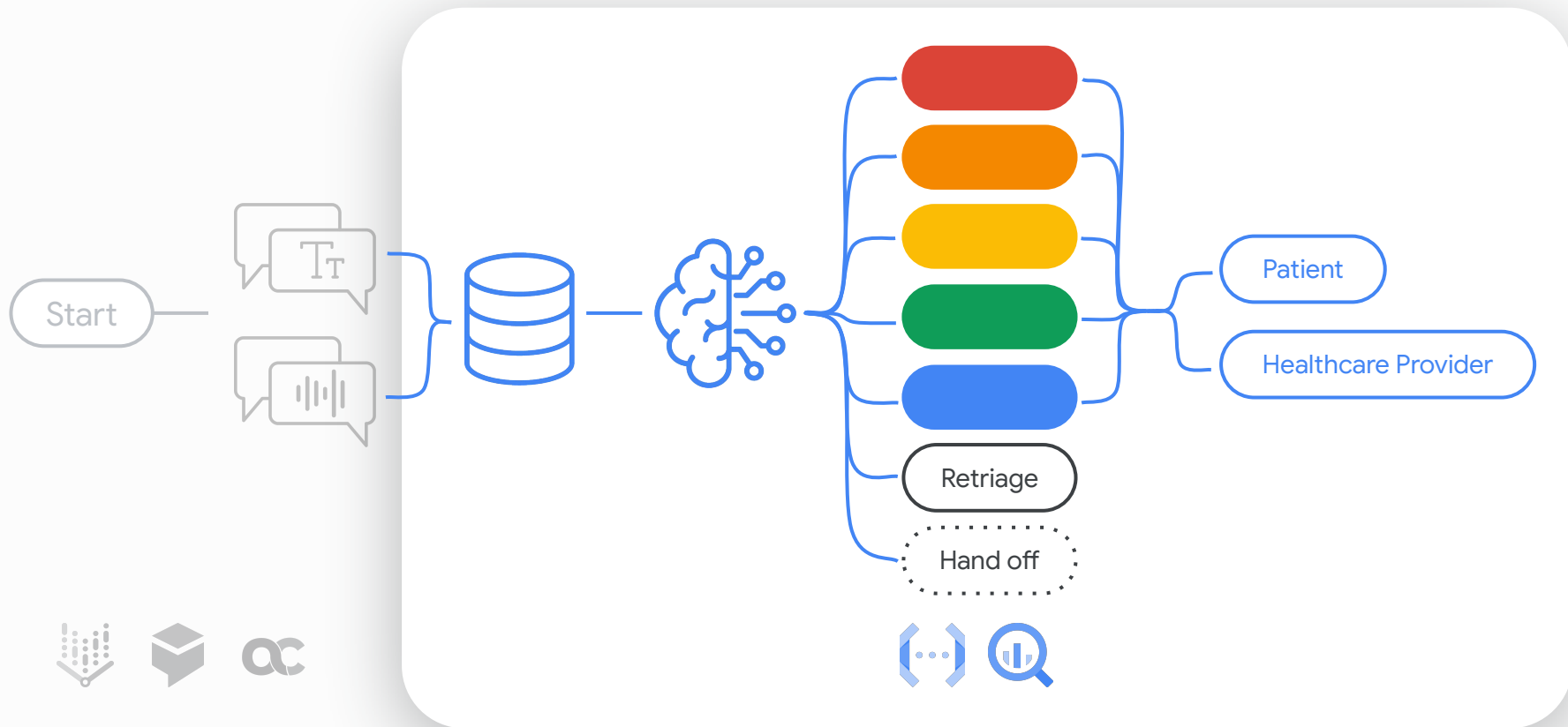
Flowchart



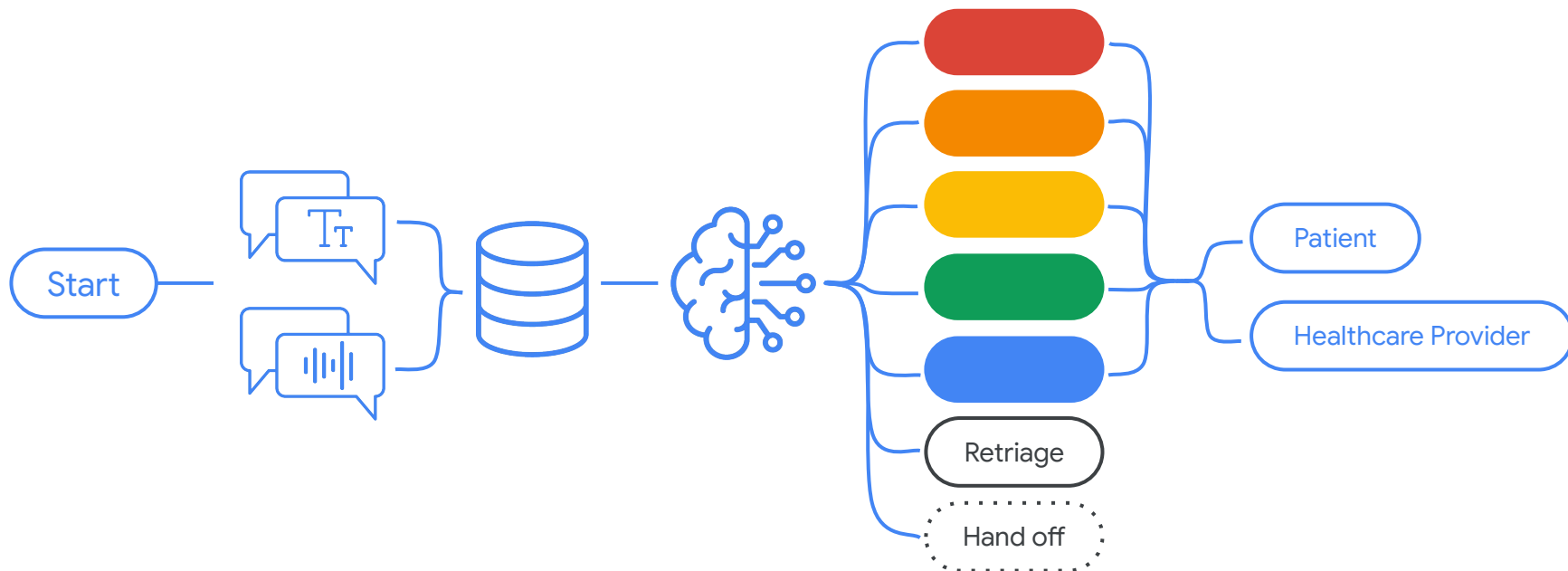
Flowchart



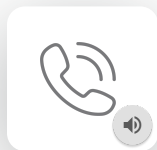
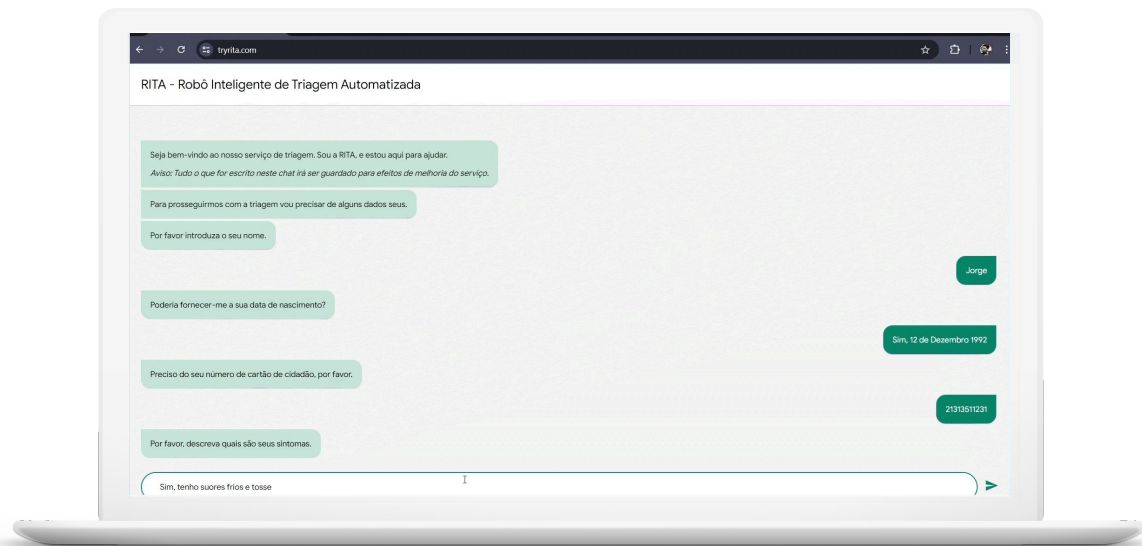
Flowchart



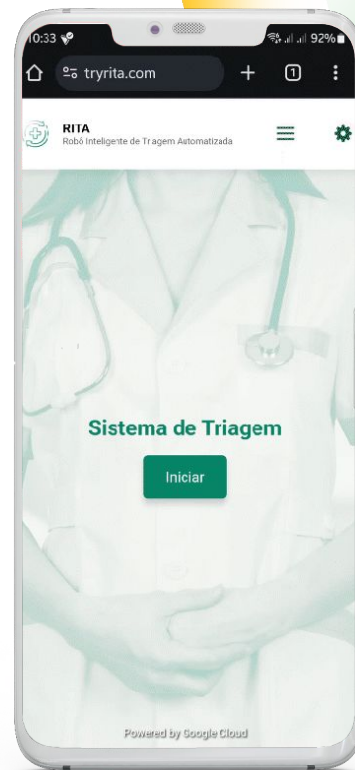
Flowchart



Prototype demo



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Now,
what's next?

Next Steps

Development

Triage dialogue flow
review

Access to **quality data**
may imply new data
collection stage

User input /
data validation
mechanisms

Patient medical history
integration

Patient **status update** in
hospital system

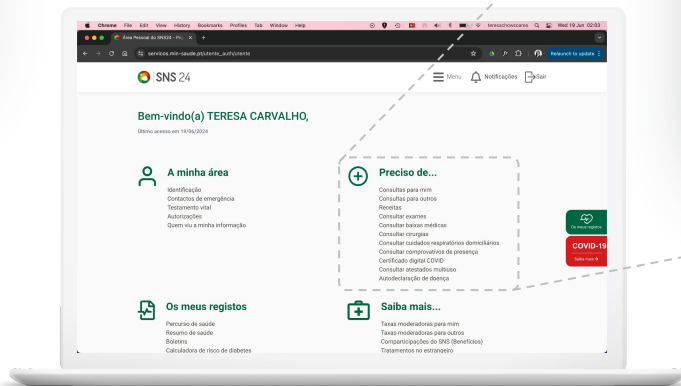
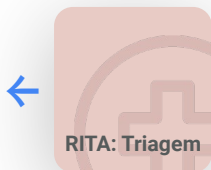
Conforming to
accessibility guidelines

Note:

Medical professionals and ethicists should be involved in the development process.

Next Steps

Integration & Launch



Preciso de...

Consultas para mim
Consultas para outros



RITA: Triagem

Receitas
Consultar exames
Consultar baixas médicas
Consultar cirurgias
Consultar cuidados respiratórios domiciliários
Consultar comprovativos de presença
Certificado digital COVID
Consultar atestados multiuso
Autodeclaração de doença

Next Steps

Integration & Launch

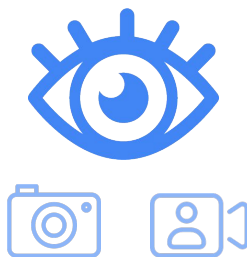


Room to Grow

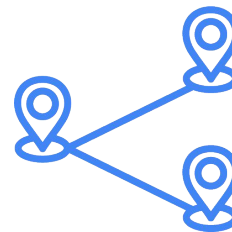
New features



monitor
physiological factors



“observe”
symptoms



suggest services
(distance & wait time)

Google Cloud

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Lisbon

Thank you!



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Q&A

